CS315 · ALGORITHMS ANALYSIS & DESIGN

Practice 2 Hash Tables

Practice on Lecture 3 · 9 questions

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01 How to solve

How to solve it in the exam

First compute `h(k)` for all the keys in one row. Then insert them one by one in the given order. With chaining, a new key goes at the end of its chain.

02 Questions

Q1 CHAINING

Insert integer keys A = [22, 1, 13, 11, 24, 33, 18, 42, 31] in order into a hash table of size 11 using the hash function h(k) = k mod 11. Collisions should be resolved via chaining, where collisions are stored at the end of a chain. Fill in the hash table after all keys have been inserted.

Index	Chain
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

Q2 LINEAR PROBING

Insert the same keys A = [22, 1, 13, 11, 24, 33, 18, 42, 31] in order into a hash table of size 11 using h(k) = k mod 11. Collisions should be resolved via linear probing. Fill in the hash table. Which key needed the largest number of probes?

0	1	2	3	4	5	6	7	8	9	10

Q3 DOUBLE HASHING

Insert keys 14, 27, 40, 19, 53, 7, 66 in order into a hash table of size N = 13 using h(k) = k mod 13. Collisions should be resolved via double hashing with d(k) = 7 − (k mod 7). Fill in the hash table after all keys have been inserted.

0	1	2	3	4	5	6	7	8	9	10	11	12

Q4 MAD · CHAINING + LINEAR PROBING

A hash table of size 11 uses the hash function h(k) = (2k + 5) mod 11. Insert keys 10, 22, 31, 4, 15, 28, 17, 88, 59 in order.

a) Resolve collisions via chaining.

b) Resolve collisions via linear probing.

Index	Chain
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

0	1	2	3	4	5	6	7	8	9	10

Q5 LECTURE EXAMPLE

Draw the 7-item hash table resulting from hashing keys 12, 44, 13, 55 and 5, using the hash function h(i) = (3i + 2) mod 7, and assume that collisions are handled by chaining.

Index	Chain
0	
1	
2	
3	
4	
5	
6	

Q6 REMOVAL · AVAILABLE

The following hash table uses h(x) = x mod 13 and linear probing:

0	1	2	3	4	5	6	7	8	9	10	11	12
		41			18	44	59	32	22	31	73	

a) Perform removeElement(44). Draw the table.

b) After the removal, which cells does findElement(31) probe? What does it return?

c) Which cells does findElement(70) probe? What does it return?

d) Perform insertItem with key 57. In which cell is it stored? Draw the table.

0	1	2	3	4	5	6	7	8	9	10	11	12

0	1	2	3	4	5	6	7	8	9	10	11	12

Q7 LOAD FACTOR

a) A hash table of size 13 stores 8 keys. Compute the load factor α and the expected number of probes for an insertion with open addressing.

b) Repeat for $\alpha = 0.9$. What do you conclude?

c) What is rehashing?

Q8 COMPRESSION MAP

A hash table uses the MAD compression map $h_2(y) = (ay + b) \bmod N$ with $N = 7$, $a = 14$ and $b = 3$.

a) Compute $h_2(2)$, $h_2(5)$ and $h_2(9)$.

b) What is wrong with this choice, and which condition does it violate?

Q9 SHORT ANSWERS

a) Discuss all available methods for handling collisions in hash tables.

b) When can you say that the hash function is "good"?

c) What is the running time of insertItem, findElement and removeElement in a log file?

d) What is the worst-case running time of the operations on a hash table, and when does it occur? What is the expected running time?

e) Which hash code map is especially suitable for strings? Why is the component sum not good for strings?

f) In double hashing, why must N be a prime, and which value can d(k) never have?